

PCB Designing Software

A reliable way to connect electronic components in a product with one another is through a printed circuit board. This article discusses some of the commonly used PCB designing software



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Australian engineer Paul Eisler invented the PCB for his radio set. Today PCBs are used in almost all electronics devices.

But designing a PCB, or even selection of a proper PCB software to design PCBs, is not so easy. One needs to know, for instance: What are the factors that should be considered while selecting the PCB software? What type of applications do you wish to address using the PCB software? Who is going to use the software—a new layout engineer or an experienced engineer? What type of circuit simulation is required—co-simulation, electromagnetic simulation or SI analysis on the PCB software? Also, budget is one of the most important criteria.

Let's look more closely at some of the most common PCB software used by the engineers, electronics enthusiasts and hobbyists.

EDWinXP

EDWinXP is a product of Visionics, and the current version available is ED-

WinXP 1.71. Joseph Abraham, manager-operations, Visionics says, "EDWinXP is a total integrated EDA/ECAD software package of seamlessly integrated, task-oriented modules, covering all stages of an electronics circuit design process—from capturing the idea of a circuit in the form of a schematic diagram to generating a full set of documentation for the manufacture and assembly of PCBs."

What's special?

EDWinXP is a complete suite that has schematics editor, PCB layout editor, autoplacers, autorouters, fabrication manager, library manager and many more. The PCB layout editor helps to design PCB layouts; the fabrication manager helps to produce the manufacturing output file, photoplotter, or NC drill. The thermal analyser helps to check the hotspots on the board. Moreover, there are more than 28,000 library components and a library editor for creative library components. Some other modules that are included in this suite are 8051, PIC, AVR and Motorola

microcontroller kit, VHDL/Truth table to diagram converter, 3D IDF file viewer, Filter designer, ODB++ export and import, VHDL editor and compiler, 3D editor and viewer, etc. The package also includes various validation tools, ensuring the correctness and integrity of the designed circuits.

Some of the features of this software are project version control, which keeps track of project versions; waveform viewer, which displays waveforms; subcircuit adapter, which allows existing SPICE sub-circuits to be converted into an EDSpice format; model parameter editors, which help in creating and managing model libraries.

According to Abraham, this software competes with PADS, OrCAD, Eagle, Altium and Allegro. Future versions will include improved 3D, routing, etc.

Users' feedback

While debugging the circuit, the software incorporates a practical automatic tracing function, which is popular with the users. Its new versions help in viewing layers better and to manage the program well. Also, the internal 'help' files are made compatible with Windows Vista.

Areas for improvement

The software comes with just 14-day trial. Users feel this is too short a period to test it. However, this period can be increased to 30 days by requesting the company.

Home page: <http://www.edwinxp.com/>
Download URL: <http://www.visionics.a.se/html/edwinxpdemo.html>

Platform supported: Windows

Kicad

This is a professional, easy-to-use, open source software for electronics circuit